

2025 MANDALAY EARTHQUAKE

RESEARCH ARTICLE

Supershear rupture sustained through a thick fault zone in the 2025 M_w 7.8 Mandalay earthquake

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On 28 March 2025, a moment magnitude (M_w) 7.8 earthquake ruptured a seismic gap along the Sagaing fault in Myanmar, generating a surface rupture 480 km in length and causing widespread damage. With holistic geodetic and seismic techniques, we resolved its rupture dynamics and fault zone structure. The rupture initiated as bilateral subshear and transitioned to supershear (~ 5.3 km/s) about 100 km south of the epicenter, sustaining this velocity for more than 200 km. The supershear segment aligns with a ~ 2 -km-thick low-velocity fault zone exhibiting $\sim 45\%$ shear wave speed reduction. We suggest that the thick fault zone, aided by fault geometry and basin structure, enabled prolonged supershear propagation. Our findings emphasize that observations and models encompassing the entire fault zone are crucial for advancing both specific-event and earthquake-cycle simulations.

The largest continental crustal earthquakes—capable of producing ruptures stretching hundreds of kilometers and affecting multiple populated centers—pose considerable seismic threats. Such events often develop into supershear ruptures in which the rupture speed exceeds crustal shear wave velocity, as reported for recent strike-slip events such as the 2001 M_w 7.8 Kokoxili earthquake (1–3), the 2002 M_w 7.9 Denali earthquake (3–5), and the 2023 M_w 7.8 Turkey earthquake (5–8). Notably, the supershear rupture portion of these events often take place on relatively simple fault geometries (1, 9, 10) or “superhighways” (11) and therefore have reduced energy dissipation into secondary fault branches, allowing the rupture to develop to supershear speed and form long surface ruptures (9). Continental strike-slip faults at plate boundaries, such as the North Anatolian (Türkiye), Sagaing (Myanmar), and San Andreas (USA) faults, are particularly concerning due to their relatively simple geometries and historical occurrence of large events.

During an earthquake rupture, part of the released energy contributes to creating damage zones or/and fault zones, characterized by diffuse inelastic coseismic deformation or substantial reduction of seismic wave speeds, as increasingly reported for active continental faults (12–19). Here, we term them coseismic damage zones when created during earthquake ruptures, primarily constrained by near-fault geodetic and field deformation observations (15, 17, 20, 21), or low-velocity fault

zones (LVFZ) when indicating the cumulative effects of historical ruptures (19, 22–26).

However, clarity is still lacking regarding the connection between these fault zones and large strike-slip ruptures, and the role of fault zones in long, sustained supershear ruptures. The recent M_w 7.8 earthquake on 28 March, 2025 in Mandalay, Myanmar, provides a valuable case to address these questions. The event took place on the Sagaing fault, which forms the eastern boundary of the Burma microplate, a sliver block created by oblique collision between the Indian Ocean and the Sunda plates (27) (Fig. 1A). The fault accommodates a tectonic slip rate of 20 to 23 mm/yr (28) (Fig. 1B). Over the past 200 years, the Sagaing fault has hosted more than 10 $M > 6.5$ earthquakes. However, a ~ 250 -km seismic gap between the cities of Mandalay and Naypyidaw had remained unruptured since at least 1897—until the 2025 M_w 7.8 event (Fig. 1C) (29). In Myanmar, the 2025 event has so far directly resulted in at least 5352 fatalities, 7108 injuries, and 538 missing individuals (30). The impacts extended beyond Myanmar, including a tall building collapse in Bangkok, Thailand, located ~ 1000 km from the epicenter (31), and substantial building damage in Yunnan province, China (32). Early reports showed that the event produced a >450 -km surface rupture and supershear rupture speed (33, 34) and also triggered seismicity in Shan-Thai plateau and southern China (35).

Through the application of multiple techniques to both seismic and geodetic datasets, we investigated the events of its surface deformation, resulting coseismic damage zone, detailed rupture process, and the associated fault zone structures. These results offer new insights into the interrelationship between supershear rupture and fault zone structure.

3D surface deformation and damage zone

To characterize surface deformation and the ruptured fault associated with the earthquake, we first analyzed multiple sets of satellite observations. We employed three independent remote sensing techniques: differential interferometric synthetic aperture radar, aperture interferometry, and pixel offset on Sentinel-1/2 and ALOS-2 satellite imagery, to obtain fourteen coseismic surface displacement images (figs. S1 and S2 and methods in the supplementary materials). These datasets, acquired from multiple viewing geometries, provide complementary sensitivity to different components of ground deformation, with an effective spatial resolution of ~ 200 m. We integrated these observations to reconstruct the full three-dimensional (3D) coseismic displacement field produced by the earthquake (fig. S3). The results reveal a rupture length of ~ 480 km, characterized by predominant right-lateral strike-slip motion (Fig. 2 and fig. S3). Vertical displacements are minor (Fig. 2C), indicating that the rupture was nearly pure strike-slip.

The coseismic surface displacements exhibit clear along-strike variations. To the first order, the north–south component of displacement (Fig. 2A) reveals a three-segment pattern. The northern segment, located north of Mandalay, shows the largest displacement for the event—up to ~ 5.5 m. Deformation was greater to the east of the fault, suggesting an east-dipping fault geometry. The central segment, extending approximately 80 km southward from Sagaing, had the smallest displacement magnitude among the three segments and near-symmetric deformation across the fault. The southern segment shows larger displacement than the middle segment but the amplitude on the western side of the fault gradually increases compared with the eastern side, implying a transition to a west-dipping fault geometry.

Fault slip and coseismic damage-zone width derived from high-resolution (10-m) Sentinel-2 displacements also reveal distinct along-strike variations (Fig. 3A and methods). North of the epicenter, sharp

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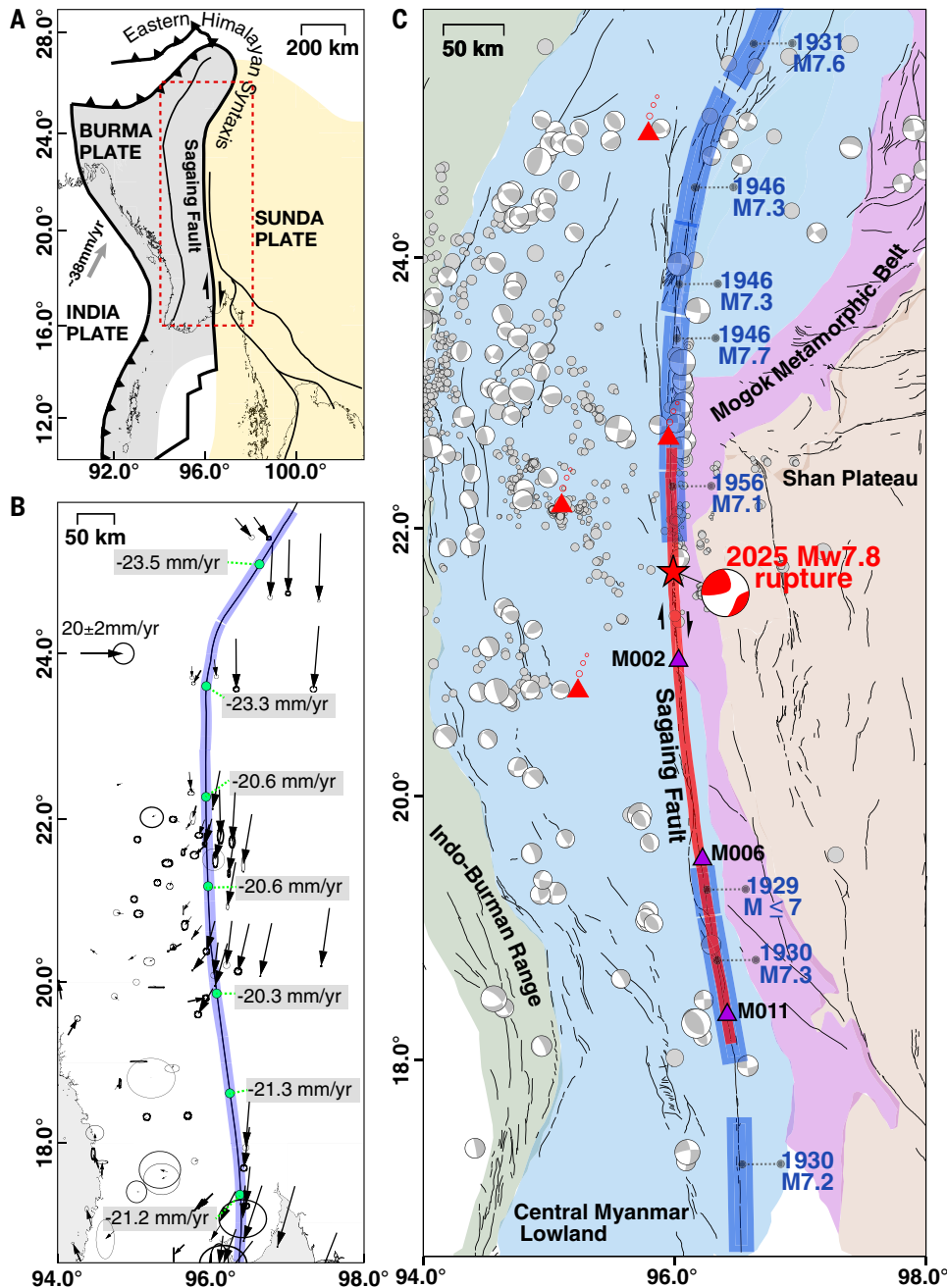


Fig. 1. Seismotectonic background. (A) Tectonic setting for the study region. The red dashed rectangle marks the locations of (B and C). (B) Slip rate distribution along the Sagaing Fault along with the GNSS observations used to derive these rates (28). GNSS velocities are relative to the Burma plate. (C) Historical earthquakes in the region. Gray beach balls represent focal mechanisms from the global centroid moment tensor (GCMT) catalog and (60). Gray dots are earthquake epicenters from the USGS catalog. Red triangles indicate locations of Quaternary volcanoes (61). The detailed fault inventory map is from (27) and the dark red line is our mapped ruptured fault trace of the 2025 event. The red star is our refined epicenter and the red beach ball is the GCMT solution.

fault-parallel offsets suggest a highly localized rupture zone with a narrow damage zone. South of the epicenter, the gradient progressively broadens across the fault, marking a transition to a wider damage zone (fig. S4). The measured fault total slip is ~1 to 6 m (Fig. 3C), distributed within a coseismic damage zone ~20 to 800 m wide (Fig. 3D). Along the fault segment north of the epicenter, where maximum coseismic slip occurred, the narrowest damage zone is ~60 m wide (Fig. 3D and fig. S4). The coseismic damage zone to the south broadens to between 90- and 190-m wide and

can be divided into three sections by thickness (Fig. 3D): ~120 m from ~20 km north to ~210 km south of the epicenter, ~90 m from ~210 to 330 km south, and ~190 m further south.

Overall, the average width of these coseismic damage zones is ~100 m (Fig. 3D), which lies at the lower end of the observed range compared with other continental strike-slip earthquakes (15, 17, 18, 36). This is due to the relatively simple geometry of the Sagaing fault, characterized by considerably fewer stepovers, bends, and branches. The segmented variation in damage zone width is more pronounced than the total slip. This contrast suggests that coseismic damage width is more strongly influenced by other factors, such as rock properties, compared with slip amplitude (18, 21). Indeed, south of the epicenter, sediment thickness increases and near-surface shear wave velocities (top 1 km) decrease (Fig. 3, D and E), as revealed by detailed shallow velocity structure imaged along the Sagaing fault (37).

Static slip model

We constrained the fault geometry and coseismic slip distribution by inverting the 3D surface deformation fields (see methods). Our preferred model resolved nine fault segments along the ~480-km rupture, achieving a variance reduction of 99.6% (figs. S5 and S6). The simple fault geometry—requiring no bifurcations or branches (black line in Fig. 3A)—contrasts with the fault branching mechanisms proposed for large strike-slip events (38). This simple geometry also deviates from the empirical relationship between rupture length and number of fault segments, as summarized for other events (39, 40), although the number of asperities may match the relationship better. Based on (39), a continental strike-slip event with a 480-km rupture length should correspond to 27 fault segments, much more than that observed for the 2025 event. Modest dip variations are evident along the strike, with the fault dipping ~70° eastward in the north, near-vertical in the central section, and slightly (~80°) west-dipping in the south (fig. S5).

The static slip model reveals that most coseismic slip is concentrated shallower than 12 km (Fig. 3B), consistent with the depth extent of interseismic locking inferred from global navigation satellite system (GNSS) observations (28). The slip distribution exhibits pronounced along-strike heterogeneity. The longest, also the largest, slip patch is located north of the epicenter and extended more than ~100 km with a peak slip of ~6 m. Southward, slip decreased to ~2.5 to 3 m along a ~70-km section, then increased again to ~4 m and remained relatively uniform over the remaining ~230 km of rupture. Superimposed on this broad pattern are short-wavelength (~10 to 20 km) variations, which are also independently detected in fault-crossing offsets from Sentinel-2 optical imagery.

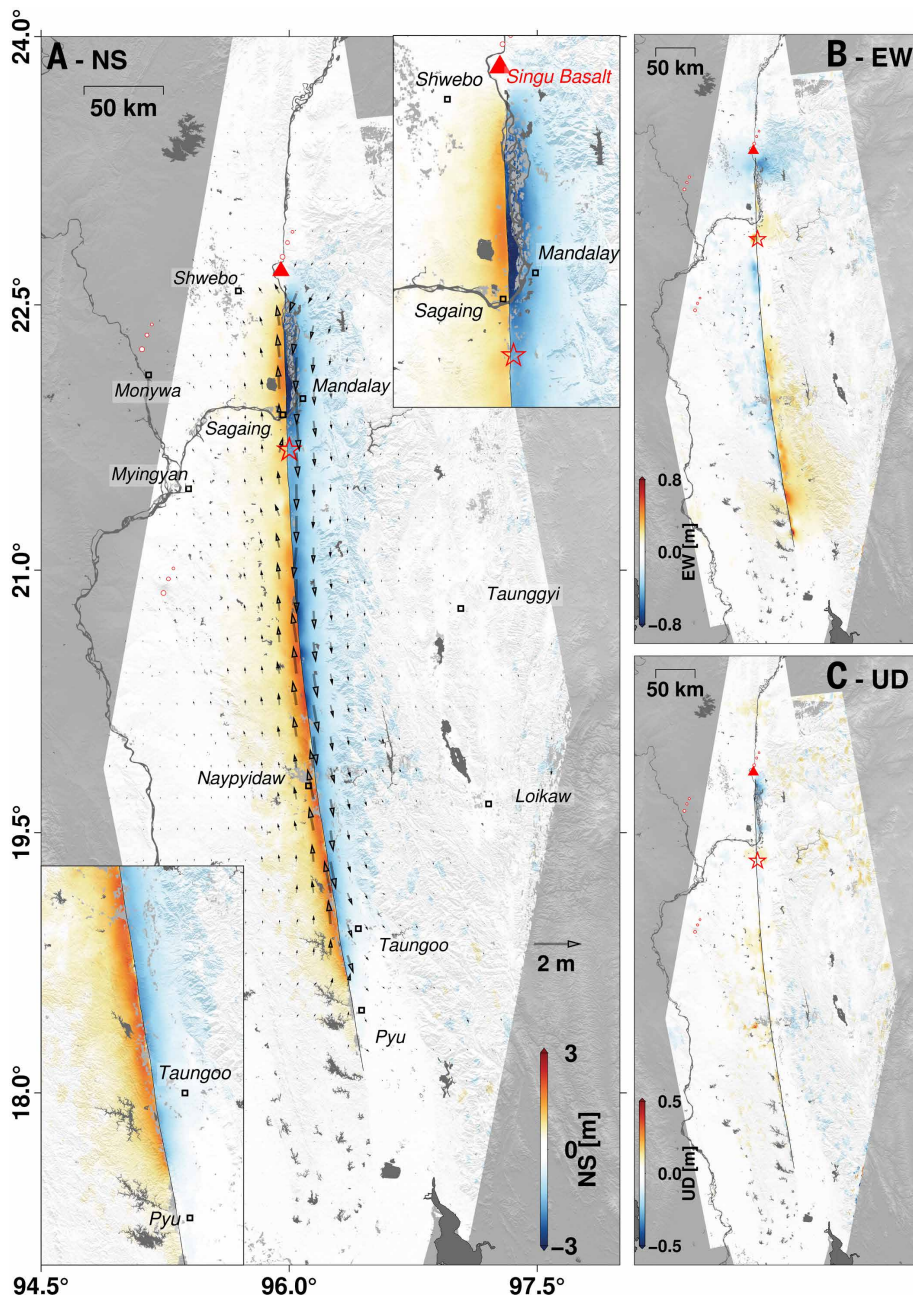


Fig. 2. 3D coseismic surface deformation map derived from satellite observations. (A) North-South direction deformation; (B) East-West; (C) up-down. The vectors in (A) represent horizontal motion and the black line is our mapped fault trace. Our refined epicenter is indicated by the red star. ALOS2 and Sentinel-1/2 images acquired from February to April 2025 were used to derive these 3D deformation maps (62) (see tables S1 and S2 for more details).

The measured total slip closely matches the near-surface slip estimated by our static slip model (Fig. 3, A and C). Checkerboard resolution tests confirmed the model's ability to resolve slip asperities across a range of spatial wavelengths (fig. S7). The averaged static stress drop estimated from our slip model was ~3 megapascals (MPa), which is consistent with that inferred for other major strike-slip faults (41).

Distinct rupture terminations were identified at both ends of the fault. To the north, the rupture terminated abruptly south of the Singu Basalt (42), coinciding with the southern limit of the 2012 M_w 6.8 Thabeikkyin earthquake rupture (Fig. 3B and fig. S8). We infer that a strong stress shadow posed by the 2012 event, combined with the

mechanical contrast between basalt and the surrounding crust, formed an effective barrier to further rupture propagation. Toward the southern end, slip amplitude and rupture depth gradually diminished, suggesting a more distributed termination process. This segment overlaps the rupture zone of the 1929 and 1930 earthquakes (Fig. 3B), although the absence of detailed historical slip models precludes further constraints. These observations explain the displacement contrast patterns shown at the northern and southern termini, shown in the insets of Fig. 2A.

Shallow slip deficit (SSD) across the rupture was overall negligible, with slip on the near-surface subfault (~1 km depth) exhibiting only ~1% deficit relative to peak slip at depth (fig. S9). This value is substantially lower than the SSD observed in previous large strike-slip earthquakes (43–45), but closer to that reported for more recent events (16). The negligible SSD is robustly resolved by the dense and near-fault geodetic coverage for the 2025 event, as also suggested by recent studies of other earthquakes (44, 46, 47). However, along-strike variations occurred: On the fault segment south of Mandalay, where the coseismic slip amplitude was less, SSD (~7%) was larger than along the other fault segments (near 0%) (fig. S9). Reanalysis of interseismic GNSS velocity fields reveals that this segment is partially creeping, with a shallow creep rate of approximately 2.3 mm/yr extending to a depth of ~9.5 km (fig. S10), in contrast to the nearly fully locked behavior of adjacent fault segments. This contrast between SSD and coseismic slip suggests that depth-dependent and laterally varying fault frictional properties contributed to observed coseismic slip variations.

Kinematic and supershear rupture

We characterized the first-order rupture evolution using a path-calibrated teleseismic back-projection (BP) (5) technique on high-frequency [(HF), 0.3 to 1.0 Hz, fig. S11] seismic data from the Australian (AU) and European (EU) arrays (fig. S12A and methods). Locations of the HF radiators from BP align closely with the fault trace (Fig. 4A). The size of the BP radiators is proportional to the power of the stacked teleseismic waveforms. Our result shows that the rupture initially propagated bilaterally, with northward and southward

segments each rupturing ~100 km at subshear speeds (~3 km/s). The southward rupture then accelerated to a supershear speed of ~5.3 km/s, maintaining this speed for at least 200 km (Fig. 4C). The robustness of the result is verified by BP applications to smaller nearby events consistent with their locations in a refined catalog (fig. S12). Synthetic tests mimicking mainshock rupture further confirmed the robustness of the result (figs. S13 and S14), which also shows that the array located away from the rupture directivity had less sensitivity (fig. S13, methods).

The evolution of rupture speed is further supported by the location and timing of subevents from a multiple point source (48) (MPS)

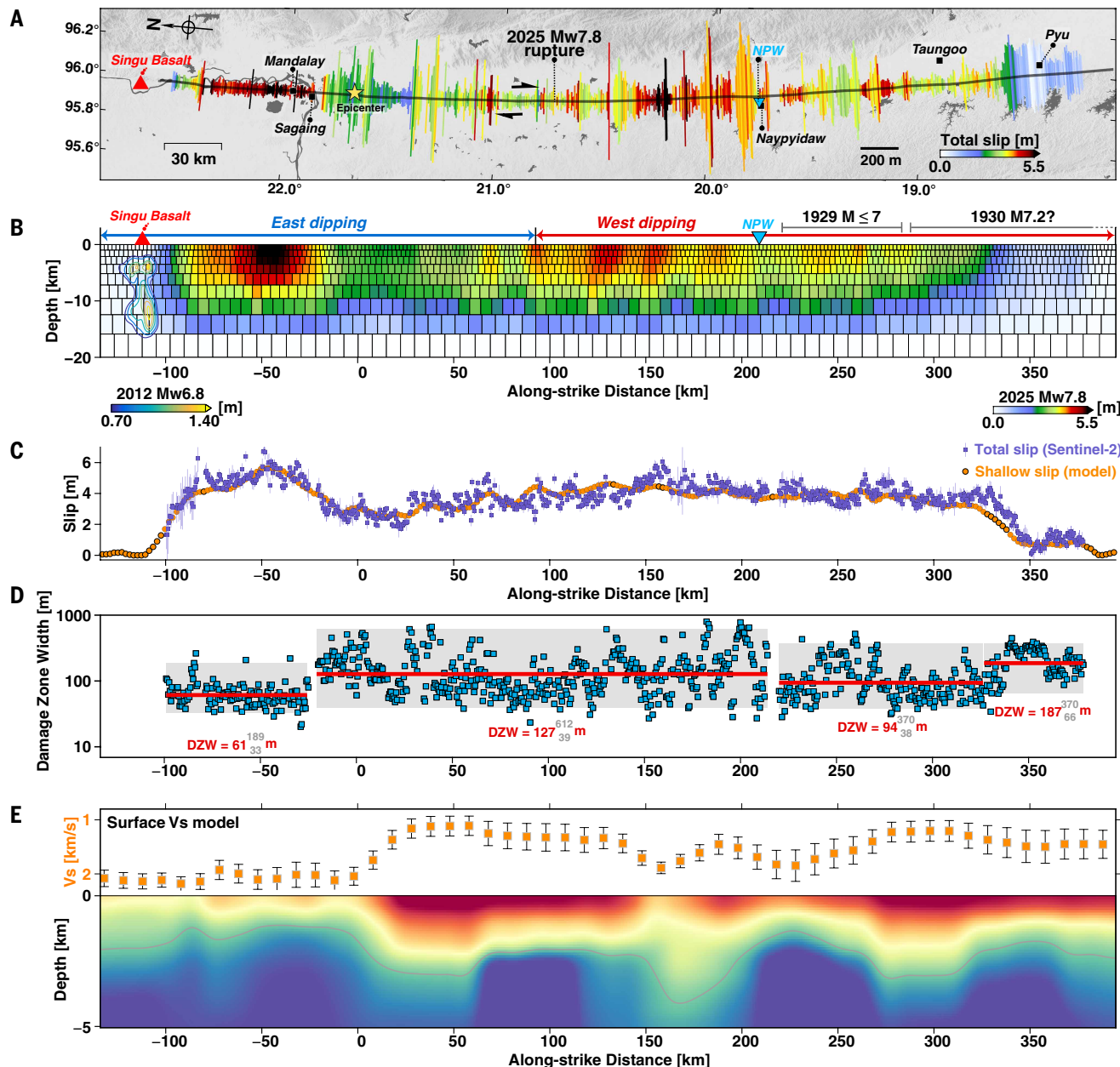


Fig. 3. Coseismic damage zone width and slip model, in comparison with velocity model. (A) Coseismic damage zone width and surface slip (62) estimated from Sentinel-2 image correlation. The length of the fault-perpendicular lines represents the width of the coseismic damage zone and the colors indicate the total slip magnitude. (B) Depth view of geodetic slip distribution for the 2025 M_w 7.8 event and the slip model of the 2012 M_w 6.8 event are shown in color-coded contours. Rupture extent for the 1929 and 1930 earthquakes is indicated at the southern end of the rupture. The purple triangle indicates the location of the NPW strong motion station. (C) Comparison between fault total slip from Sentinel-2 displacement [purple squares with uncertainty defined as standard deviation (one-sigma) between fault-parallel displacement (10 m binned mean) and fitting displacement] and shallow (≤ 1 km depth) slip amplitudes in the static slip model (orange dots). (D) Damage zone width (blue squares). The red lines are the averaged damage zone width of four segments. (E) Shear wave speed model in the top 5 km from (37). Orange squares represent the numerical average of uppermost 1-km surface shear wave velocity and uncertainty, in which the uncertainty is defined in (37). The gray line in (E) indicates the sedimentary base.

inversion using local-to-regional strong motion waveform and teleseismic body waves (Fig. 4B, fig. S16, and methods). MPS results fit the observations closely, with an average waveform cross-correlation coefficient of 0.8 across all components and stations (fig. S17). The result consists of 10 subevents (fig. S18), and the spatiotemporal evolution of these subevents shows that the mainshock began with a bilateral rupture that transitioned to supershear propagation ~ 100 km south of the epicenter (Fig. 4C). The overall source time function captures the low-frequency component of the moment-rate function derived from the finite fault inversion (Fig. 4D, inset; see details below).

Strong-motion waveforms recorded at the Naypyidaw (NPW) station (49), located 2.6 km from the fault trace at 19.779°N (blue triangle in Fig. 4A), provide independent evidence for supershear rupture propagation. To ensure the reliability of the absolute timing and waveform data from this station, we verified the accuracy of its GNSS time synchronization by comparing teleseismic arrival times with those recorded globally (fig. S15 and methods).

We then refined the mainshock epicenter and origin time using a nearby, well-located calibration event (fig. S19 and methods). Our analysis yields a revised epicenter at the latitude of 21.641°N , ~ 37 km

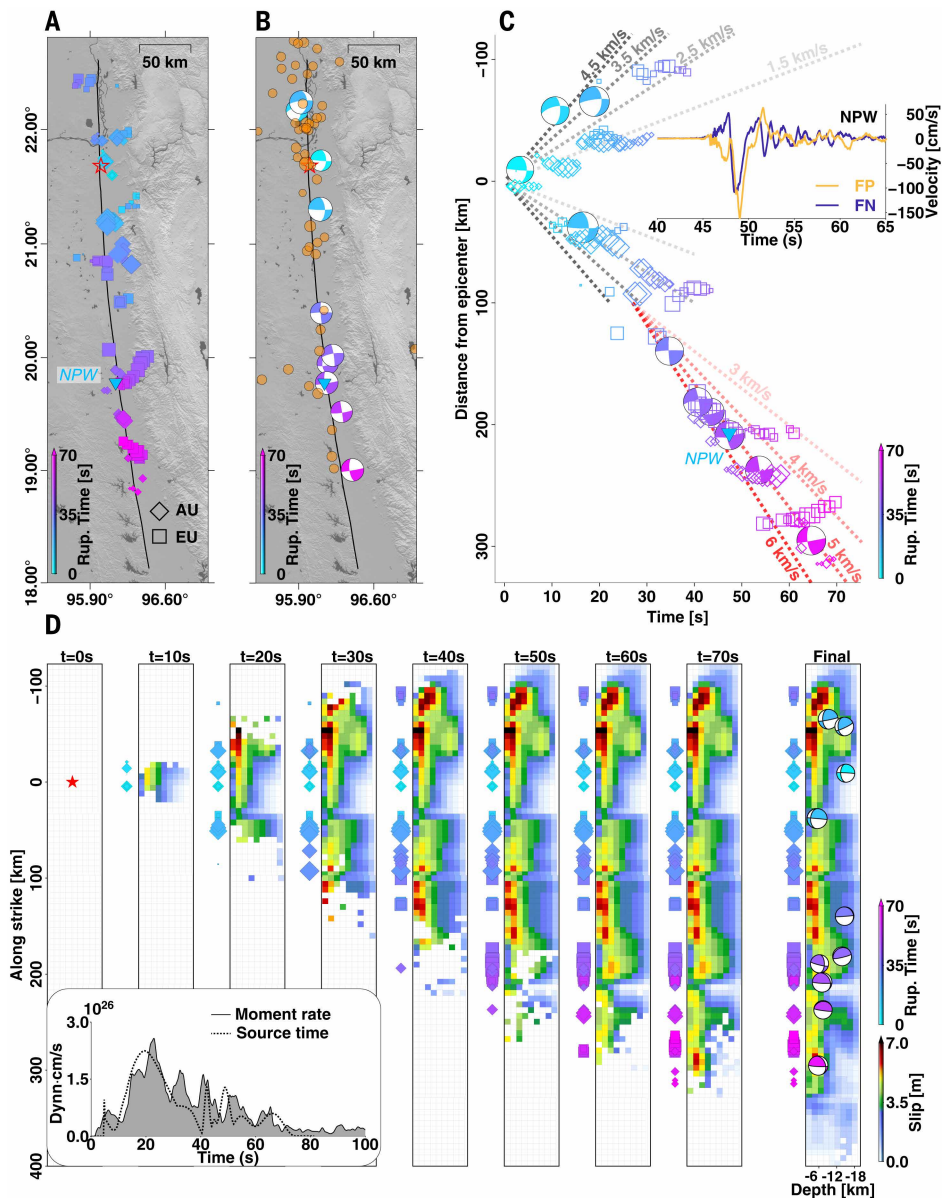


Fig. 4. Kinematic rupture models. (A) Map view of the back-projection (BP) results (62) from the EU array (squares) and the AU array (diamonds), color coded by time and sized by the power of stacked teleseismic waveforms. (B) MPS results (62) in map view, in which beach balls are color-coded by peak time in the source time function, projected to the low hemisphere. Orange dots indicate aftershocks provided by the USGS catalog. The blue triangles in (A) and (B) represent the location of the NPW station. (C) Rupture speed analysis and horizontal component velocity waveform at the NPW station. Fault-parallel (FP) and fault-normal (FN) directions are relative to the fault azimuth angle of 173 degrees. (D) Finite fault model with time (10 s) (62). Inset shows the moment rate function of the finite fault model (gray shaded area) and the total source time function of the MPS solution, with focal mechanism solutions projected to the western hemisphere.

south of that in the initial United States Geological Survey (USGS) report (22.013°N) (<https://www.usgs.gov/programs/earthquake-hazards/earthquakes>), but in closer agreement with the Thailand catalog (21.68°N). The refined origin time (2025-03-28 06:20:55.209 coordinated universal time) is 2.5 s later than the USGS report, again aligning more closely with the Thailand catalog (<https://www.tmd.go.th/en/Earthquake>).

The supershear rupture pulse recorded at the NPW station, identified as the onset of the largest amplitude in the velocity waveform, arrived at $\sim 48 \pm 0.5$ s (Fig. 4C inset). Given the station's distance of

206 km from the refined epicenter—closely approximating the rupture distance along the fault—the inferred average rupture speed was ~ 4.3 km/s, exceeding the local shear-wave velocity and thus indicating supershear propagation. Notably, the rupture phase arrival time at NPW closely matches the rupture timing derived from BP and MPS results at the same location (blue triangle in Fig. 4C). Further evidence for supershear rupture propagation is provided by the NPW waveform characteristics. Theoretical asymptotic analyses and laboratory rock experiments (50) demonstrate that when the Mach cone front generated by a sustained supershear rupture sweeps a near-fault station, the resulting ground motion is dominated by fault-parallel (FP) particle velocity relative to the fault-normal (FN) component. This pattern is clearly observed at the NPW station (Fig. 4C, inset, and fig. S20A). Moreover, the observed FP-to-FN velocity amplitude ratio suggests a rupture velocity of 5.0 to 5.3 km/s (fig. S20B and methods), consistent with estimates from BP and MPS.

Using the refined epicenter and rupture speed constraints from the preceding analyses, we performed a finite fault inversion that jointly fits teleseismic body waves, near-fault strong-motion records, and the 3D surface displacement field (methods). The resulting finite fault model (Fig. 4D) provides excellent fits to all datasets (figs. S21 to S23) and yields a moment-rate function consistent with that derived from the MPS source time function (Fig. 4D, inset), lending high confidence to key features of the model. These include the fault-geometry transition from east-dipping in the north to west-dipping in the south, the spatial distribution of slip, and the supershear rupture velocity observed along the central segment of the rupture. Collectively, multiple independent analyses and inversions corroborate a sustained supershear rupture propagating southward from ~ 100 km to at least 310 km along the strike from the epicenter (Fig. 4).

Thick LVFZ for supershear segment

We used passive seismic measurements to investigate the influence of the internal structure of the Sagaing fault zone. Along the supershear rupture segment of the 2025

event, three broadband seismic stations were located within 1 km of the surface rupture (Fig. 5A and fig. S24). Teleseismic P-waves recorded at these stations exhibited major time shifts between the vertical and radial components (fig. S24A). Notably, earthquakes originating from the western side of the fault show larger time shifts than those from the eastern side, indicating a clear back-azimuthal variation (fig. S24B). These time shifts resulted in delayed arrivals in receiver function (RF) waveforms, and when the shift exceeded a threshold (e.g., >1 s), a negative phase was observed at the zero time (Fig. 5B, figs. S25 to S27, and methods).

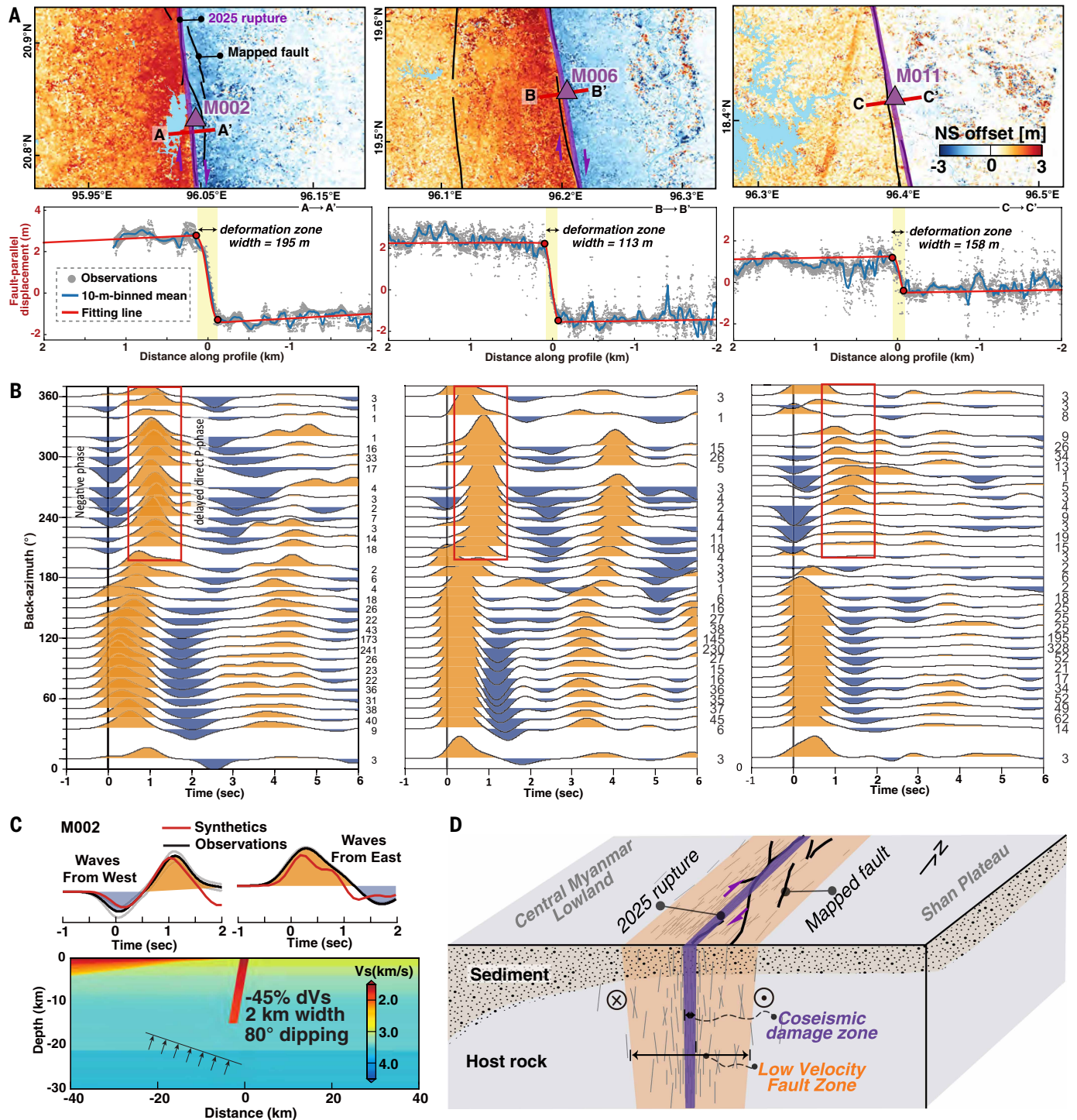


Fig. 5. Relationship between the coseismic damage zone and the low-velocity fault zone sustaining supershear rupture. (A) North-South displacement and across-fault profiles at the three broadband stations (M002, M006, and M011; Fig. 1C). Black and purple lines represent mapped faults and the 2025 rupture. **(B)** Stacked RF waveforms at M002, M006, and M011, grouped by the back-azimuth of incoming waves, with the number of stacked traces indicated on the right side of each panel. A clear transition is observed from a positive to negative zero-lag phase at back-azimuths near 180°. **(C)** 2D synthetic modeling results illustrating the effect of a dipping low-velocity fault zone on RF waveforms. A west-dipping fault zone produces a negative zero-lag phase for waves coming from the west, while maintaining a positive first P-phase for waves from the east, consistent with the observations. **(D)** Schematic illustration showing the relationship between the coseismic damage zone and the low-velocity fault zone.

These negative zero-lag phases in RFs cannot be explained by a 1D layered velocity model and are often neglected in conventional RF studies (51). Here, we argue that the negative zero-lag phase represents true structural signals caused by 2D effects of low-velocity fault zone structure. Through synthetic tests we show that basin models (37) are unable to reproduce the negative phase at time zero (fig. S28B). The negative zero-lag phase occurs only within specific back-azimuth ranges (Fig. 5B), which rules out sensor misorientation. It is observed when a different frequency band (fig. S27A) and deconvolution method (fig. S27B) are used. Therefore, the signal is not caused by sidelobe effects during filtering nor is it an artifact of RF calculation. The transition from a positive to a negative zero-lag phase occurred at back-azimuths near 180°—consistent with the polarity reversal observed in Fig. 5B—and closely aligns with the strike of the Sagaing fault, supporting an interpretation of a west-dipping fault zone structure.

We modeled the observed polarity reversal of the zero-lag phase with a west-dipping low-velocity fault zone structure (Fig. 5C and methods). A grid search was performed on the dipping angle, fault zone width, and velocity reduction, to find the best combined fit to observed RF waveforms. The velocity reduction ranged from 25 to 45% with 10% intervals; fault zone width varied from 1 to 3 km with 0.5 km intervals; and the dip angle ranged from 60° to 90° with 10° intervals. For each grid point, we calculated the amplitude ratio between the negative phase near time zero and the subsequent positive phase, then computed the misfit between observed and synthetic waveforms. The search shows that a west-dipping fault zone with a dip angle of 70° to 80°, a thickness of ~2 km, and a shear-wave velocity reduction of approximately 45% is needed to achieve the negative zero-lag phase and corresponding amplitude (figs. S28 and S29). The grid search results also indicate an uncertainty of ~0.5 km in fault width and ~10% in velocity reduction (fig. S30 and methods). The inferred west-dipping fault geometry is also consistent with that derived from geodetic inversions (fig. S31). The ~2-km width of the fault zone exceeds those of other well-studied faults such as the San Andreas, Ridgecrest, Landers, and Calico faults (table S3), although the observed velocity reduction is comparable (fig. S32 and methods).

The drivers of supershear rupture

The simple fault geometry and thick fault zone probably provide the most essential conditions behind the supershear and subshear-to-supershear transition distance of the 2025 event. A fault zone produces strong head waves that travel faster than the initial rupture front, creating dynamic fault stress perturbations that promote the development of a new supershear rupture front, as confirmed by dynamic earthquake simulations (12, 52, 53). Fault zones need to be sufficiently thick to facilitate a supershear transition. Damic rupture simulations show that the critical fault zone width should be at least 2 to 3 times the critical nucleation size of earthquakes $L_c = \frac{GD_c}{\Delta\tau_s}$, where G is shear modulus, D_c is the critical slip distance in slip-weakening friction law, and $\Delta\tau_s$ is the fault strength drop, i.e., the difference between fault static and dynamic shear strength. Assuming $G = 30$ GPa, $D_c = 1$ m, and $\Delta\tau_s = 50$ MPa (methods), and yielding a L_c of 600 m, the observed Sagaing fault zone width would be ~3 to 4 L_c . Such thick fault zones allow the transition to supershear rupture even with low background stress. Given the estimated static stress drop of 3 MPa, the relative stress drop—defined by the ratio of static stress drop to the strength drop—is likely <0.1 in this region, substantially lower than the relative stress drop needed for supershear transition in homogeneous media (12). Another feature of supershear rupture in thick fault zones is the increase in the supershear transition length with the fault zone width (12). A relative stress drop of <0.1 and a fault zone width of 3 to 4 L_c indicate that the transition length is >100 L_c (12), which matches the ~100-km transition length observed from the seismological analysis.

Other factors may also have facilitated the supershear rupture transition in the 2025 event. The observed supershear transition occurred near an along-strike fault-geometry transition, where the fault dip changes from eastward to subvertical (slightly west-dipping, Fig. 3B and fig. S5). Dynamic simulations demonstrated that geometrical kinks, such as dip changes or bends, can act as stress concentrators that temporarily slow and then reaccelerate the rupture, favoring supershear transition once the accumulated energy is released (54). As the rupture propagated onto the west dipping fault, the smooth fault geometry provided the “superhighway” to help sustain the supershear rupture, similar to what may have happened for other supershear events (9, 11). The relatively thick (~2 to 3 km) sedimentary basin along the southern fault segment (Fig. 3E) serves as another favorable condition to sustain supershear rupture, as the waves reflected from the bottom of the basin could have increased shear stress on the fault and promoted rupture propagation (55, 56).

In contrast to the relatively simple surface rupture of the 2025 event, fault traces mapped by earthquake geologists reveal secondary fault branches and bifurcations (Figs. 5D and 1C). These deviations from the straight 2025 rupture likely reflect past earthquakes that created more complex geomorphological signatures (27). The considerable fault zone thickness (~2 km) further suggests that ruptures in the past may have involved both the main fault and secondary strands. Observations indicate that fault zones may take hundreds or thousands of years for faults to heal from coseismic damages of large events (57). Dynamic simulations also show that coseismic rupture can produce off-fault plastic deformation that leads to permanent damage in the fault zone (58). Therefore, cumulative fault damage from previous events, including events similar to the 2025 event, have likely formed the observed thick fault zone (Fig. 5D).

Our observations and models reconcile the coseismic fault geometry, supershear rupture, and fault damage zone in a region where supershear rupture may have been governed by the cumulative imprint of past earthquakes. This inherited complexity appears to have guided the rupture dynamics for the 2025 M_w 7.8 Mandalay event. The Sagaing fault thus exemplifies how long-term fault zone evolution may influence earthquake behavior, a pattern that might apply to other plate boundary-type strike-slip faults. Specifically, supershear rupture along these faults may be more likely to occur where the fault zones are wide and fault geometry is simple. These findings highlight the need for both specific-event and earthquake-cycle models to incorporate more comprehensive and accurate fault zone structures (59), which may critically control rupture mode, speed, and size across successive earthquake generations. The results also provide new insight into assessing the potential hazard of major earthquakes along continental faults with recognized seismic gaps, such as those associated with the southern San Andreas fault in CA and the North Anatolian fault near Istanbul, Türkiye.

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SUPPLEMENTARY MATERIALS

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